Yulia Volkova

yulia-volkova.github.io
LinkedIn • yuulia.volkova@gmail.com

I am a MATS 9.1 scholar working on AI control evaluations with Rhys Ward and Redwood Research. I have 7+ years of experience building insider threat detection and communication surveillance systems at Behavox for Tier-1 financial institutions. I first hand witnessed AI integration into the regulation sector, which inspired me to move and work on AI safety. Further details can be found on my website.

Selected Publications

2026 — Control Evaluations for Covert Malicious Fine-Tuning. *Submitted to NeurIPS 2026*
Buckworth, T., Moreno, R., Panda, A., **Volkova, Y.** Supervised by Dr Francis Rhys Ward.

2026 — Commitment To Cooperation With Self-Negotiated Contracts *COLM 2026*
Wyse, T., Bustos, K., **Volkova, Y.**, Kleiman-Weiner, M.
Algoverse AI Safety Research Fellowship, Cooperative AI track. Mentor: Max Kleiman-Weiner.

2025 — ParaScopes: What do Language Models Activations Encode About Future Text?
Pochinkov, N., **Volkova, Y.**, Vasileva, A., Chereddy, S. V. R.
Algoverse AI Safety Research Fellowship. PI: Callum McDougall, mentor: Nicky P.

Research Programs

Jan 2026 – Present — MATS 9.0

- Control evaluations for covert malicious fine-tuning under Francis Rhys Ward, advised by Tyler Tracy and James Lucassen (Redwood Research)
- Selected for MATS 9.0 Spotlight Talk

Nov 2025 — Athena Women in AI Safety Research Fellowship

- Mechanistic interpretability approach to monitoring chain-of-thought faithfulness under Arun Jose

Oct 2025 — SPAR Fellowship

- CoT monitorability project under Qiyao Wei (MATS 8.0, Cambridge University)

Aug 2025 — Algoverse AI Safety Research Fellowship

- Mechanistic interpretability under Nicky P
- Cooperative AI under Max Kleiman-Weiner

Jul 2025 — ARBOx2 Oxford AI Safety Bootcamp

- Completed the ARENA curriculum

Work Experience

BEHAVOX

Behavox is a company specialising in AI-native compliance and conduct surveillance for financial institutions.

***2020 – 2025* — ML Production Teamlead**

As a lead my key responsibilities include identifying the product scope, development of multilingual scalable NLP models, deploying applications to production, cross-team communication and roadmapping, continuous product improvement given production feedback

Automation and optimization achievements

- Created an automated process that improved development time of ML models from 3 days to 10 minutes, including tests, documentation, UI controls, and app logic built in one click.
- Modernised the QA process with a synthetic data generator, automatically generating and uploading data to AWS S3 for use in test environments via a Jenkins pipeline.
- Consolidated 4 legacy security monitoring products into 1 solution, enabling it to scale and freeing up backend capacity previously required to support them.

***2017 – 2020* — Data Scientist**

Key responsibilities: development of multilingual applications and reports to flag statistical deviations and compliance breaches in client data, designing and building infrastructure to support application creation, QA and maintenance

- Landed proof of concepts that resulted in 3+ years client contracts.
- Improved compliance model recalls from 70% to 90%.
- Reduced client bug requests from 1/day to 1/month by setting up an efficient QA process.
- Learned Java to remove the Analytics dependency on Backend teams.
- Developed best practices for code quality as one of the first Data Science hires and created foundational documentation that helped onboard 13 people.

Education

London School of Economics
MSc Economics

- Distinction in dissertation (dissertation in behavioural economics)
- Courses included Advanced Macroeconomics, Econometrics, Quantitative Methods

London School of Economics
BSc Philosophy & Economics

- Distinction
- Courses included Mathematical Methods, Statistics, Econometrics, Philosophy of Science

Media & Talks

- **Aug 2026 — GovAI expert panel** — "OpenAI x Hugging Face incident: What happened? What's next?" (event)
- **Jun 2026 — MATS Project Talks at LISA and Google DeepMind London** — presented the covert malicious fine-tuning project (F. R. Ward's stream) to research audiences at both offices
- **Apr 2026 — MATS 9.0 Spotlight Talk** — opened the London event (YouTube)
- **Mar 2026 — FellowCast Podcast at Constellation** — "Covert Malicious Fine-Tuning" (Spotify)

Skills Summary

Technology	Groovy, Java, Python, Pandas, Matplotlib, Seaborn, Scikit-learn, NumPy, PyTorch, SQLAlchemy, MySQL, Hbase, Kotlin, R, Stata, AWS S3, unit testing, code review, CI/CD, Apache Spark
Data Science	Multivariate classification, SVM, logistic regression, regularization, multicollinearity, feature selection, cross validation, class balancing, PCA, gradient boosting, t-SNE representations, KNN, decision trees. Performance metrics: recall, precision, F-score, ROC-AUC, PR-AUC
Statistics	Hypothesis testing, A/B testing, asymptotic theory, regression analysis, vector auto-regressions, time series analysis, panel data analysis: fixed and random effects, IV estimator method

Software & frameworks	Jira, TeamCity, Jenkins, GitLab, Kibana, IntelliJ, PyCharm, Bitbucket, Github, Jupyter Labs, FastAPI, AirFlow
Leadership	Team planning, mentoring, interviewing and hiring, and roadmapping
Other Interests	Oil painting, skiing, hiking